AED Intertidal Soil Model

Technical Manual

Part A: Soil Moisture Component

Part B: Soil Temperature and Surface Energy Balance

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Aquatic Ecodynamics

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Part A

Soil Moisture Component

Part A: Soil Moisture Component

1. Introduction

This document describes the soil moisture component of the AED intertidal soil model. The model simulates vertical water movement, storage and loss in a soil column subject to periodic tidal submergence and atmospheric drying. Three coupled sub-models handle different aspects of the moisture system:

1. **Matrix flow** – implicit finite-difference solution of the Campbell (1985) infiltration equation for the unsaturated zone.
2. **Water table tracking** – external position tracker that converts net fluxes at the base of the unsaturated zone into water table movement via specific yield.
3. **Macropore (dual-domain) flow** – optional preferential flow pathway representing cracks and structural pores, coupled to the matrix solver and water table.

The model operates on a vertically discretised soil column with geometric layer spacing (fine near the surface, coarsening with depth). At each timestep the model evaluates whether the soil surface is submerged by comparing the external water level against the surface elevation, then applies the appropriate boundary conditions.

2. Single-Domain Matrix Flow

2.1 Governing equations

Vertical moisture redistribution in the matrix follows the Campbell (1985) formulation. The one-dimensional continuity equation for liquid water in a soil column is:

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z} - S$$

where θ is volumetric water content ($m^3\ m^{-3}$), q is the vertical water flux ($m\ s^{-1}$, positive downward), z is depth below the surface (m), and S is a sink term (root water uptake).

The flux q is the sum of a matric (pressure-driven) component and a gravitational component:

$$q = -K(\psi)\left(\frac{\partial \psi}{\partial z} - g\right)$$

where ψ is the matric potential (J kg^{-1}, negative in unsaturated soil), K is the hydraulic conductivity (kg s m^{-3}), and $g = 9.8$ m s^{-2}.

2.2 Hydraulic functions

The Campbell (1985) power-law relationships link moisture content to matric potential and hydraulic conductivity:

$$\psi = \psi_e \left(\frac{\theta_s}{\theta} \right)^b$$

$$K = K_s \left(\frac{\psi_e}{\psi} \right)^n, \qquad n = 2 + \frac{3}{b}$$

where: - ψ_e is the air-entry potential (J kg^{-1}, negative), the matric potential at which air first enters the largest pores during drainage - θ_s is the saturated water content (porosity) - K_s is the saturated hydraulic conductivity (kg s m^{-3}) - b is the Campbell exponent, which controls the shape of the retention curve (larger b = finer-textured soil with stronger water retention)

Porosity is derived from bulk density:

$$\theta_s = 1 - \frac{\rho_b}{\rho_m}$$

where ρ_b is the soil bulk density (Mg m^{-3}) and ρ_m is the mineral density (default 2.6 Mg m^{-3}).

2.3 Vapour transport

A simple vapour diffusion flux is included between adjacent nodes, driven by the gradient in relative humidity:

$$q_v = \frac{0.66\, D_v\, \rho_v \left(\theta_s - \bar{\theta} \right)}{\Delta z} \left(h_{i+1} - h_i \right)$$

where $D_v = 2.4 \times 10^{-5}$ m^2 s^{-1} is the binary diffusion coefficient of water vapour in air, $\rho_v = 0.017$ kg m^{-3} is a reference vapour density, $\bar{\theta}$ is the mean water content of adjacent nodes, and $h_i = \exp\left(M_w \psi_i / RT_i \right)$ is the equilibrium relative humidity at node i, with $M_w = 0.018$ kg mol^{-1} and $R = 8.31$ J mol^{-1} K^{-1}.

The factor $\left(\theta_s - \bar{\theta} \right)$ represents the air-filled porosity through which vapour can diffuse; vapour transport is therefore negligible when the soil is near saturation.

2.4 Evapotranspiration

Total potential evapotranspiration is partitioned into soil evaporation and transpiration by the leaf area index:

$$E_p = e^{-0.82\,\mathrm{LAI}} \cdot \mathrm{ET}_p, \qquad T_p = \mathrm{ET}_p - E_p$$

Surface evaporation is driven by the humidity gradient between the top soil node and the atmosphere:

$$E_{surf} = E_p \cdot \frac{h_2 - h_a}{1 - h_a}$$

where h_2 is the relative humidity at the surface node and h_a is the atmospheric relative humidity.

Transpiration is distributed across root-containing layers via a resistance network. Root extraction from each layer depends on the soil-to-leaf water potential gradient divided by the sum of soil, root, and leaf resistances.

When a constant ET rate is prescribed via `fixed_et`, the model applies a top-down extraction scheme after the solver step. Water is removed layer by layer from the surface downward until the demand is met or the residual water content (0.02 m^3 m^{-3}) is reached. In the presence of a coupled water table, the ET demand is split between unsaturated extraction and direct water table evaporation (Section 3.4).

3. Water Table Tracking

3.1 Rationale

The Campbell (1985) solver represents matric potential as a negative quantity and cannot represent the positive pore pressures that exist below the water table. The solver is therefore restricted to the unsaturated zone above the water table. Water table position is tracked externally by converting net fluxes at the base of the unsaturated column into vertical movement via the specific yield.

3.2 Truncated solver domain

At each timestep the model identifies the number of *active* (unsaturated) layers above the water table. A layer is active if its top is above the water table depth. The layer straddling the water table (top above, bottom below) is included; `truncated_view()` creates a copy of the soil parameter object containing only the active layers, with the bottom boundary repositioned at the water table depth. The Campbell solver runs on this truncated column with a `fixed_head` lower boundary condition representing the saturated zone.

Layers below the water table are held at full porosity and are not modified by the solver.

3.3 Field-capacity cap on newly exposed layers

When the water table drops and exposes a formerly saturated layer, that layer's moisture is set to the field capacity:

$$\theta_{fc} = \theta_s - S_y$$

rather than leaving it at full porosity. This prevents the solver from recycling the specific yield water back into the water table on the next timestep (the solver would see a fully saturated layer, compute downward drainage flux, and the water table tracker would interpret this as recharge, raising the water table back up).

The field-capacity cap is applied inside `_solve_moisture_wt()` by comparing the current count of active layers (`n_active`) against the previous timestep's count (`_prev_n_active`). Any layer whose index falls between the two values is

a newly exposed layer.

3.4 Water table water balance

The water table position is updated each timestep from the net flux entering or leaving the saturated zone. The update equation is:

$$\Delta z_{wt} = -\frac{F_{net}}{S_y}$$

where Δz_{wt} is the change in water table depth (positive = deepening), F_{net} is the net flux into the water table (m of water, positive = recharge), and S_y is the specific yield.

The net flux comprises up to four terms:

Term	Symbol	Sign convention	Description
(a) Drainage	F_d	+ into WT	Gravity-driven percolation from the unsaturated zone
(b) Capillary rise	F_c	- from WT	Solver pulls water upward from the fixed-head BC
(c) Direct WT evaporation	E_{wt}	- from WT	Evaporation extracted directly from the saturated zone
(d) Macropore drain	F_m	+ into WT	Cascade water exiting the base of the macropore column

Terms (a) and (b) are not separated explicitly; they are computed together from a column mass balance or from the solver's boundary flux, depending on the step type.

3.5 Wet-step vs dry-step strategy

The method for computing the flux from the water table differs between submerged (wet) and exposed (dry) timesteps:

Wet steps use the solver's Darcy flux at the lower boundary (`bottom_flux`):

$$F_{net} = F_{bot} + F_m + F_{cap_overflow}$$

where F_{bot} is the solver's bottom boundary flux, F_m is macropore bottom drain, and $F_{cap_overflow}$ is any post-solver matrix cap overflow (Section 4.5). Ponding infiltration and capillary rise both contribute to `bottom_flux` and cannot be separated, but the sign is consistently positive (net downward) under ponding conditions.

Dry steps use a column mass balance to determine how much water the solver exchanged with the water table:

$$F_{from_wt} = \left(W_{after} - W_{before}\right) + E_{unsat}$$

$$F_{net} = -F_{from_wt} - E_{wt} + F_m$$

where W_{before} and W_{after} are the total water stored in the unsaturated zone before and after the solver step (m), and E_{unsat} is the actual unsaturated-zone evaporation this step (m). The quantity F_{from_wt} is positive when the solver pulled water upward from the water table (capillary rise exceeds drainage). Computing W_{before} after the field-capacity cap (Section 3.3) is essential; otherwise the Sy release from newly exposed layers would be misinterpreted as capillary rise.

The mass-balance approach is preferred for dry steps because it avoids a subtle recycling artefact: the solver's bottom flux can include both real drainage and numerical adjustments from the Dirichlet boundary, and under dry conditions these can interact with the matrix cap (Section 4.5) to create spurious fluxes.

3.6 Direct water table evaporation

When the water table is shallow, evaporation can extract water directly from the saturated zone via capillary connectivity to the surface. The fraction of the total ET demand taken directly from the water table is:

$$f_{direct} = \exp\left(-\frac{5\, z_{wt}}{d_{ext}}\right)$$

where z_{wt} is the water table depth (m) and d_{ext} is the extinction depth (m, default 0.10 m). The constant 5 is chosen so that $f_{direct} \approx 0.007$ at $z_{wt} = d_{ext}$.

The ET demand is then split:

$$E_{wt} = E_{demand} \cdot f_{direct}, \qquad E_{unsat} = E_{demand} \cdot \left(1 - f_{direct}\right)$$

Direct water table evaporation is subtracted from the water table reservoir and does not pass through the unsaturated-zone solver.

4. Dual-Domain (Macropore) Model

4.1 Concept

Intertidal soils frequently develop structural cracks, root channels, and faunal burrows that create preferential flow paths. These macropores transmit water much faster than the soil matrix under saturated or near-saturated conditions, but contribute negligibly to storage or capillary processes.

The model represents this using a dual-domain approach. Total porosity is partitioned into two domains:

$$\theta_s = \theta_{matrix_cap} + f_{macro}$$

where f_{macro} is the macropore porosity fraction and $\theta_{matrix_cap} = \theta_s - f_{macro}$ is the maximum water content of the matrix domain. The two domains occupy the same physical space but have independent moisture states and different flow physics.

The macropore domain is activated by setting `f_macro > 0` in the soil parameters.

4.2 Macropore gravity cascade

Water movement in the macropore domain is modelled as a simple top-down gravity cascade (kinematic wave), without capillary or diffusive terms. During ponding, water enters the macropore column from the surface; during dry periods no external input occurs but stored macropore water continues to drain.

Surface input (ponding only):

$$q_{in} = \min\left(k_{macro} \cdot f_{macro} \cdot \Delta t, \quad V_{empty}\right)$$

where k_{macro} is the macropore hydraulic conductivity (m s^{-1}) and V_{empty} is the total empty macropore volume in the active column:

$$V_{empty} = \sum_{i=0}^{n_{active}-1} \max\left(f_{macro} - \theta_{macro,i},\ 0\right) \cdot \Delta z_i$$

Capping the input at V_{empty} prevents overfilling the macropore column.

Layer-by-layer cascade: Starting from the surface, each layer receives the incoming flux, drains under gravity, and passes any overflow and drainage to the layer below:

$$\theta_{macro,i} \leftarrow \theta_{macro,i} + \frac{q_{in}}{\Delta z_i}$$

If $\theta_{macro,i} > f_{macro}$, the excess overflows:

$$Q_{overflow} = \left(\theta_{macro,i} - f_{macro}\right) \cdot \Delta z_i, \qquad \theta_{macro,i} = f_{macro}$$

Gravity drainage from the layer is:

$$Q_{drain} = \min\left(\theta_{macro,i},\ \frac{k_{macro} \cdot f_{macro} \cdot \Delta t}{\Delta z_i}\right) \cdot \Delta z_i$$

The input to the next layer is the sum of drainage and overflow:

$$q_{in,i+1} = Q_{drain} + Q_{overflow}$$

Water exiting the base of the active macropore column is returned as `bottom_drain`, which represents direct recharge to the water table.

4.3 Matrix-macropore exchange

After the cascade, water exchanges between the macropore and matrix domains driven by the difference in relative saturation:

$$X_i = \alpha \cdot f_{macro} \cdot \left(S_{r,macro} - S_{r,matrix}\right) \cdot \Delta t$$

where α is the exchange rate coefficient (s^{-1}), and:

$$S_{r,macro} = \frac{\theta_{macro,i}}{f_{macro}}, \qquad S_{r,matrix} = \frac{\theta_{matrix,i}}{\theta_{matrix_cap}}$$

The exchange is limited to prevent either domain from exceeding its capacity or going negative:

- If $X_i > 0$ (macro to matrix): $X_i \leq \theta_{macro,i}$ and $X_i \leq \theta_{matrix_cap} - \theta_{matrix,i}$

- If $X_i < 0$ (matrix to macro): $|X_i| \leq 0.5 \cdot \theta_{matrix,i}$ and $|X_i| \leq f_{macro} - \theta_{macro,i}$

The exchange rate α has a timescale of roughly $1/\alpha$ seconds. For $\alpha = 5 \times 10^{-5}$ s $^{-1}$ this is approximately 5.5 hours.

4.4 Coupling with the unsaturated-zone solver

The Campbell solver is unaware of the dual-domain split. It uses the full porosity θ_s as the saturation limit for the matrix. This is a deliberate design choice: modifying the solver's porosity would require changes throughout the tridiagonal system and could affect convergence. Instead, the model enforces the matrix capacity constraint externally.

The per-step sequence in the model orchestrator is:

1. **Macropore step** – run cascade and exchange on the unsaturated layers only (0 to `n_active - 1`)
2. **Add exchange to matrix** – update matrix moisture with the exchange fluxes
3. **Matrix cap enforcement** – clip or redistribute any matrix moisture above θ_{matrix_cap} (method depends on step type; see Section 4.5)
4. **Campbell solver** – solve the matrix flow on the truncated unsaturated column
5. **Post-solver cap handling** – handle any solver-introduced excess above cap
6. **ET extraction** – remove water from top layers (dry steps only)
7. **Water table update** – compute net flux, update water table position
8. **Zero saturated macropores** – set macropore moisture to zero in layers below the water table

4.5 Matrix cap enforcement: wet vs dry steps

The solver can push matrix moisture above θ_{matrix_cap} because it treats full porosity as the saturation limit. How this excess is handled depends critically on the step type:

Wet steps use `_enforce_matrix_cap()`, which redistributes excess water back into the macropore domain:

1. For each unsaturated layer where $\theta_{matrix,i} > \theta_{matrix_cap}$:
   - Transfer the excess to the macropore domain: $\theta_{macro,i} \leftarrow \theta_{macro,i} + \text{excess}$
   - If the macropore is also full, route the remaining overflow to `macro_drain` (direct water table recharge)
2. This is applied both before and after the solver. Post-solver overflow is counted in the water table recharge.

This redistribution is physically justified during ponding: the solver fills the surface layers to porosity via the Dirichlet boundary condition, and the excess above the matrix cap represents water that should occupy the macropore domain.

Dry steps use a simple clip:

$$\theta_{matrix,i} = \text{clip}\left(\theta_{matrix,i},\ 10^{-7},\ \theta_{matrix_cap}\right)$$

No water is routed to the macropore domain. This is essential to prevent a recycling artefact:

1. The solver pulls water upward from the water table via capillary rise (fixed-head BC)
2. Capillary rise fills matrix layers near the water table above the matrix cap
3. If `_enforce_matrix_cap` routes the excess to macropores, the cascade drains it back to the water table
4. The water table rises, the solver pulls more water – an infinite loop

By clipping without routing, the small amount of excess (which represents solver-introduced water in the macropore pore space, where capillary rise should not physically occur) is absorbed. The post-solver clip is applied before computing W_{after} so the mass balance correctly reflects that this water was never truly extracted from the water table.

4.6 Below-water-table macropore moisture

Below the water table, all pore space (matrix and macropore) is part of the saturated zone. The water table tracker accounts for this storage via the specific yield. To prevent double-counting, macropore moisture is set to zero in all layers below the water table at the end of each timestep (`_zero_saturated_macropores`).

5. Vertical Discretisation

5.1 Layer geometry

The soil column is divided into n layers with geometric spacing: thin layers near the surface resolve the steep moisture and temperature gradients driven by boundary-condition switching, while thicker layers at depth reduce computational cost.

Layer thicknesses follow a geometric series:

$$\Delta z_i = \Delta z_0 \cdot r^{i-1}, \qquad i = 1, \ldots, n$$

where $\Delta z_0 = 0.005$ m (5 mm) is the thinnest surface layer and r is a growth ratio solved by bisection so that the sum of all layer thicknesses equals the total column depth (minus a 1 mm surface offset):

$$\sum_{i=1}^{n} \Delta z_0 \cdot r^{i-1} = D_{max} - 0.001$$

For $n = 20$ layers and $D_{max} = 1.0$ m, the growth ratio is approximately 1.205, giving a surface layer of 5 mm and a bottom layer of about 17.5 cm.

5.2 Depth array indexing

The layer depths array has $n + 2$ entries (indices 0 to $n + 1$):

Index	Depth	Meaning
0	0.0	Ground surface
1	0.001	Top of first layer (1 mm offset)
2	0.006	Bottom of first layer / top of second

Index	Depth	Meaning
…	…	…
$n+1$	D_{max}	Bottom of last layer

The Campbell solver uses an internal Fortran-style 1-based grid with phantom boundaries. The mapping between the external 0-based layer index j and the internal node index i is: $i = j + 2$, for $j = 0, ..., n-1$.

6. Boundary Conditions

6.1 Submergence test

At each timestep the model compares the external water level h_w against the soil surface elevation z_s:

- $h_w \geq z_s$: **submerged** (wet step)
- $h_w < z_s$: **exposed** (dry step)

6.2 Surface boundary conditions

Submerged (wet): - Ponding Dirichlet BC: the surface node is pinned at saturation potential ($\psi = \psi_e$) inside the solver's convergence loop. This drives infiltration into the column. - No evaporation.

Exposed (dry): - The surface is open to the atmosphere. Evaporation is driven by the humidity gradient between the surface node and the air (Section 2.4). - The solver receives zero ET; extraction is handled externally via `_apply_et()` after the solver step.

6.3 Lower boundary conditions

When water table tracking is enabled (the standard configuration), the Campbell solver always uses `fixed_head` at the water table depth. The solver sees a saturated boundary that can supply water (capillary rise) or accept drainage.

Without water table tracking, three options are available: - `free_drain`: phantom deep boundary, gravity-driven drainage exits the column - `fixed_head`: saturated boundary at the column base - `no_flux`: sealed base, allowing an internal water table to develop

7. Timestep Sequence

Each model timestep proceeds as follows:

7.1 Wet step (`_step_wet`)

1. If dual-domain: run macropore cascade with `ponding=True`, apply exchange to matrix, enforce matrix cap (with routing to macropores/WT)
2. Run Campbell solver on the truncated unsaturated column with ponding BC
3. If dual-domain: enforce matrix cap on solver output (overflow counted as WT recharge)

4. Update water table: $F_{net} = F_{bot} + F_m + F_{cap_overflow}$
5. If dual-domain: zero macropore moisture below new water table
6. Solve temperature

7.2 Dry step (`_step_dry`)

1. Solve temperature
2. If dual-domain: run macropore cascade with `ponding=False`, apply exchange to matrix, clip matrix to cap (no routing)
3. Compute ET demand; split into unsaturated and direct WT components
4. Run Campbell solver on the truncated unsaturated column (no ponding, no ET)
5. Apply unsaturated ET extraction top-down
6. If dual-domain: clip solver output to matrix cap
7. Compute W_{after} and column mass balance
8. Update water table: $F_{net} = -F_{from_wt} - E_{wt} + F_m$
9. If dual-domain: zero macropore moisture below new water table

8. User Guide: Parameters and Configuration

8.1 Creating a model

```
        from intertidal_soil import SoilParams,
IntertidalSoilModel, TidalForcing

        params = SoilParams(
            n_layers=20,            # number of vertical
layers
            max_depth=1.0,          # total column depth (m)
            psi_e=-2.0,             # air-entry potential
(J/kg)
            K_s=5e-4,               # saturated hydraulic
conductivity (kg s/m^3)
            b=3.0,                  # Campbell 'b' exponent
            bulk_density=1.5,       # soil bulk density
(Mg/m^3)
            Sy=0.2,                 # specific yield
            # Dual-domain (set f_macro=0.0 or omit for single-
domain):
            f_macro=0.03,           # macropore porosity
fraction
            k_macro=0.01,           # macropore conductivity
(m/s)
            alpha_exchange=5e-5,    # exchange rate (1/s)
        )

        model = IntertidalSoilModel(
            params,
            surface_elevation=0.5,  # soil surface elevation
(m, datum-referenced)
            dt=900.0,               # timestep (seconds)
            water_table_depth=0.9,  # initial WT depth below
surface (m)
```

```
            fixed_et=0.010,          # constant ET rate
(m/day), or None for PET calc
            evap_extinction_depth=0.10,  # direct WT evap
extinction depth (m)
        )
```

8.2 Soil hydraulic parameters

Parameter	Symbol	Units	Typical range	Effect
`psi_e`	ψ_e	J kg^{-1}	-0.5 to -5.0	More negative = finer soil, stronger suction at air entry
`K_s`	K_s	kg s m^{-3}	10^{-5} to 10^{-2}	Higher = faster drainage and infiltration
`b`	b	–	2 to 12	Higher = stronger water retention, steeper retention curve
`bulk_density`	ρ_b	Mg m^{-3}	1.1 to 1.7	Determines porosity via $1 - \rho_b/\rho_m$
`Sy`	S_y	–	0.05 to 0.30	Drainable fraction; controls WT response speed

Coarse sands: low $|\psi_e|$, high K_s, low b. Clay soils: high $|\psi_e|$, low K_s, high b.

8.3 Water table parameters

Parameter	Units	Description
`water_table_depth`	m	Initial depth below surface. Set to `None` to disable WT tracking.
`Sy`	–	Specific yield. Determines how far the WT moves per unit recharge.

Parameter	Units	Description
`evap_extinction_depth`	m	Depth below which direct WT evaporation is negligible. Default 0.10 m. Larger values allow the WT to supply evaporation from deeper positions.

8.4 Macropore parameters

Parameter	Symbol	Units	Typical range	Effect
`f_macro`	f_{macro}	–	0.01 to 0.05	Fraction of total porosity allocated to macropores. Set 0 to disable.
`k_macro`	k_{macro}	m s^{-1}	10^{-3} to 10^{-1}	Macropore drainage speed. Much faster than matrix K_s.
`alpha_exchange`	α	s^{-1}	10^{-6} to 10^{-3}	Exchange rate. Timescale ~ $1/\alpha$ seconds.

Macropore effects are most pronounced when: - `k_macro >> K_s` (preferential flow is much faster than matrix flow) - `alpha_exchange` is moderate (slow enough that macropore and matrix moisture are out of equilibrium during a tidal cycle, fast enough that exchange occurs within hours) - `f_macro` is small relative to porosity (macropores are a small fraction of total pore space, consistent with field observations of 1–5%)

8.5 Evapotranspiration options

Parameter	Units	Description
`fixed_et`	m day^{-1}	Prescribed constant ET rate. Overrides the built-in PET calculation.
`evap_scale`	–	Multiplier on the built-in PET (only used when `fixed_et` is None). Set 0 to disable ET.
`lai`	–	Leaf area index. Partitions ET between soil evaporation and transpiration. Low LAI (e.g. 0.1 for bare mud) means most ET is direct soil evaporation.

The built-in PET model is intentionally simple: $\mathrm{PET} = T_{air}/3500$ (m day^{-1}) when rain is below 5 mm day^{-1}. For realistic applications, users should provide `fixed_et` from an external energy-balance or Penman-Monteith calculation.

8.6 Forcing data

The model requires a `TidalForcing` object with five time-series arrays of equal length:

Field	Units	Description
`time`	s	Time from simulation start
`water_level`	m	Water surface elevation (same datum as `surface_elevation`)
`air_temp`	degC	Air temperature
`water_temp`	degC	Water column temperature
`rain`	m day^{-1}	Rainfall rate

The forcing is linearly interpolated to the model timestep internally. To represent an exposed (dry) period, set `water_level` below `surface_elevation` (e.g. -1.0 m).

8.7 Running a simulation

```
        state = model.initialise(
            air_temp=22.0,         # initial/boundary air
temperature (degC)
            deep_temp=18.0,        # deep boundary temperature
(degC)
            init_moisture=0.15,    # initial unsaturated VWC
            spin_up_days=5.0,      # temperature spin-up
duration
        )

        output = model.run(state, forcing)
        results = output.to_arrays()
```

The results dictionary contains:

Key	Shape	Units	Description
`time`	(n_steps,)	s	Timestep times
`moisture`	(n_steps, n_layers)	m^3 m^{-3}	Matrix volumetric water content
`macro_moisture`	(n_steps, n_layers)	m^3 m^{-3}	Macropore water content (dual-domain only)
`temperature`	(n_steps, n_layers)	degC	Soil temperature
`water_table_depth`	(n_steps,)	m	Water table depth below surface

Key	Shape	Units	Description
`evap`	(n_steps,)	m	Total evaporation this timestep
`recharge`	(n_steps,)	m	Net WT flux this timestep (positive = recharge)
`direct_wt_evap`	(n_steps,)	m	Direct WT evaporation component
`f_direct`	(n_steps,)	–	Fraction of ET from the saturated zone
`is_submerged`	(n_steps,)	bool	Whether the surface was submerged

9. Known Limitations

1. **Single soil type per column.** All layers share the same hydraulic parameters (ψ_e, K_s, b). Layered soil profiles (e.g. sand over clay) are not supported.

2. **No lateral flow.** The model is strictly one-dimensional (vertical). Lateral seepage, groundwater inflow from adjacent areas, and tidal aquifer dynamics are not represented.

3. **No hysteresis.** The same retention curve is used for wetting and drying. In reality, the drying curve retains more water at a given suction than the wetting curve.

4. **Solver uses full porosity in dual-domain mode.** The Campbell solver does not know about the macropore/matrix split and treats porosity as the saturation limit. The matrix cap is enforced externally, requiring careful handling to avoid mass-balance artefacts (Section 4.5).

5. **No macropore capillary processes.** The macropore domain is purely gravitational. No capillary rise, no diffusion, no root uptake from macropores. This is physically reasonable for large structural pores but less so for mesopores.

6. **Simplified PET.** The built-in evaporation model is a rough temperature-based approximation. For quantitative applications, an externally computed ET rate should be supplied via `fixed_et`.

7. **No salt or solute transport.** The model tracks water only. Salinity effects on soil hydraulic properties (osmotic potential, flocculation) are not included.

8. **Specific yield is constant.** S_y does not vary with depth or water table position. In reality, specific yield depends on the retention properties of the soil at the water table depth.

Part B

Soil Temperature and Surface Energy Balance

Part B: Soil Temperature and Surface Energy Balance

1. Introduction

This document describes the soil temperature component of the AED intertidal soil model. The model simulates vertical heat conduction through a soil column that is periodically submerged by tidal water and exposed to the atmosphere. Two coupled sub-models handle different aspects of the thermal system:

1. **Subsurface heat conduction** – implicit backward-Euler finite-difference solution of the one-dimensional heat equation, with moisture-dependent thermal properties.
2. **Surface energy balance (SEB)** – Newton-iteration solver that determines the surface temperature of exposed soil from the balance of radiative, turbulent, and ground heat fluxes.

When the soil surface is submerged, the temperature boundary condition comes directly from the hydrodynamic model's water temperature. When exposed, the SEB computes a physically-based surface temperature that reflects the interplay between solar heating, longwave radiation, turbulent exchange with the atmosphere, evaporative cooling, and conduction into the soil.

The temperature model couples to the moisture model in both directions: soil moisture controls thermal conductivity and heat capacity (via the Johansen model), and the surface energy balance determines the latent heat flux that drives evaporation.

2. Subsurface Heat Conduction

2.1 Governing equation

Vertical heat conduction in the soil follows Fourier's law. The one-dimensional heat equation for a soil column is:

$$C(\theta)\frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left[\lambda(\theta)\frac{\partial T}{\partial z}\right]$$

where T is temperature (K or degC), z is depth below the surface (m), $C(\theta)$ is the volumetric heat capacity (J m^{-3} K^{-1}), $\lambda(\theta)$ is the thermal conductivity (W m^{-1} K^{-1}), and θ is the volumetric water content (m^3 m^{-3}).

Both C and λ are functions of soil moisture, coupling the thermal and hydraulic models.

2.2 Numerical discretisation

The soil column is divided into n layers with geometric spacing (fine near the surface, coarsening with depth). The layer boundaries are defined by the depth array $d[0..n+1]$ where $d[0] = 0$ (surface) and $d[n+1]$ is the column base.

Layer thicknesses and centre depths:

$$\Delta z_i = d[i+1] - d[i], \quad z_{c,i} = \frac{d[i] + d[i+1]}{2}, \quad i = 0, \ldots, n-1$$

Inter-layer distances and interface conductivities (harmonic mean for internal interfaces):

$$\Delta z_{c,i} = z_{c,i} - z_{c,i-1}, \quad \lambda_{i+\frac{1}{2}} = \frac{2\lambda_i \lambda_{i-1}}{\lambda_i + \lambda_{i-1}}$$

The top interface uses the top-layer conductivity: $\lambda_{-\frac{1}{2}} = \lambda_0$, with distance from surface to first cell centre: $\Delta z_{c,0} = z_{c,0}$.

2.3 Implicit time integration

An implicit backward-Euler scheme is used, yielding a tridiagonal system that is unconditionally stable regardless of timestep size or layer thickness:

$$C_i \Delta z_i \frac{T_i^{n+1} - T_i^n}{\Delta t} = \frac{\lambda_{i-\frac{1}{2}}}{\Delta z_{c,i}} \left(T_{i-1}^{n+1} - T_i^{n+1} \right) + \frac{\lambda_{i+\frac{1}{2}}}{\Delta z_{c,i+1}} \left(T_{i+1}^{n+1} - T_i^{n+1} \right)$$

This is solved by the Thomas algorithm (tridiagonal matrix algorithm) at each timestep.

2.4 Boundary conditions

Top boundary – Dirichlet (standard): The surface temperature T_s is prescribed:

$$T_{-1}^{n+1} = T_s$$

This enters the top-layer equation as a known forcing term on the right-hand side. When the soil is exposed to the atmosphere, T_s is determined by the surface energy balance (Section 4), which iteratively solves for the temperature at which radiative, turbulent, and conductive fluxes balance. When the soil is submerged, T_s is taken directly from the hydrodynamic model's water temperature (Section 6). In both cases, the heat solver receives a temperature, not a flux – the ground heat flux G is a diagnostic output (Section 2.5), not an input to the solver.

Top boundary – Neumann (optional): A surface heat flux G (W m^{-2}, positive into soil) can in principle be prescribed directly as a source term in the top-layer equation. This mode exists in the code (`surface_flux` parameter in `soil_temp()`) but is **not used** because the thin surface layer (~3 mm) produces numerical oscillations under a flux BC (see Section 8.1 for details). The Dirichlet approach is unconditionally stable and is the only mode used in practice.

Bottom boundary – Dirichlet: A fixed deep temperature T_{deep} is prescribed at the base of the column. For tropical coastal settings, $T_{deep} \approx 26$ degC is typical.

2.5 Heat flux diagnostic

After solving for the new temperature profile, the surface heat flux into the soil is computed diagnostically:

$$G = \frac{\lambda_0}{\Delta z_{c,0}} \left(T_s - T_0^{n+1} \right)$$

This value is reported but does not feed back into the solver (the solver used T_s as Dirichlet, not G as Neumann).

2.6 Temperature spin-up

The `initial_temp()` function creates a quasi-equilibrium temperature profile by running the heat equation repeatedly with constant surface and deep boundary temperatures for a specified spin-up period (default 30 days at 15-minute timesteps). The initial profile is a linear interpolation from T_{air} to T_{deep}.

3. Thermal Properties

3.1 Thermal conductivity – Johansen model

The effective thermal conductivity λ_{eff} interpolates between dry and saturated conductivities using the Kersten number K_e:

$$\lambda_{eff} = \lambda_{dry} + \left(\lambda_{sat} - \lambda_{dry} \right) \cdot K_e$$

Kersten number (normalised conductivity, empirical):

$$K_e = \log_{10} \left(S_r \right) + 1.0, \quad K_e \in [0, 1]$$

where $S_r = \theta / \phi$ is the degree of saturation and ϕ is porosity.

Dry conductivity (Johansen 1975):

$$\lambda_{dry} = \frac{0.135 \rho_b + 64.7}{\rho_s - 0.947 \rho_b}$$

where ρ_b is bulk density and ρ_s is mineral (particle) density, both in kg m^{-3}.

Saturated conductivity (geometric mean):

$$\lambda_{sat} = \lambda_m^{(1-\phi)} \cdot \lambda_w^{\phi}$$

Parameter	Symbol	Value	Units
Mineral conductivity	λ_m	2.5	W m^{-1} K^{-1}
Water conductivity	λ_w	0.57	W m^{-1} K^{-1}
Mineral density	ρ_s	2600	kg m^{-3}

For a typical Giralia sandy-clay ($\rho_b = 1550$ kg m^{-3}, $\phi = 0.404$): $\lambda_{dry} \approx 0.27$ W m^{-1} K^{-1}, $\lambda_{sat} \approx 1.3$ W m^{-1} K^{-1} – a factor of ~5 variation with moisture content.

3.2 Volumetric heat capacity

A simple volume-weighted mixture:

$$C = (1-\phi)\,C_m + \theta\,C_w + (\phi-\theta)\,C_a$$

Component	Symbol	Value	Units
Mineral	C_m	2.0×10^6	J m^{-3} K^{-1}
Water	C_w	4.18×10^6	J m^{-3} K^{-1}
Air	C_a	1.25×10^3	J m^{-3} K^{-1}

The air contribution is negligible; the dominant effect is that wet soil has roughly double the heat capacity of dry soil ($C_{dry} \approx 1.2 \times 10^6$, $C_{sat} \approx 2.9 \times 10^6$ J m^{-3} K^{-1}).

3.3 Implications for intertidal soils

The strong moisture dependence of both λ and C produces qualitatively different thermal behaviour in wet and dry soil:

- **Wet soil** (after tidal inundation): high λ, high C. Heat from the warm water boundary penetrates deeply into the profile. Temperature changes are slow and smooth.
- **Dry soil** (prolonged exposure): low λ, low C. The surface responds rapidly to radiative forcing (large diurnal amplitude, peak temps 50–60 degC) but heat is confined to a thin surface layer – the soil is an effective insulator.

4. Surface Energy Balance

4.1 Overview

When the soil surface is exposed to the atmosphere, a surface energy balance determines the surface temperature T_s. The balance equation is:

$$SW_{net} + LW_{down} - LW_{out} - H - LE = G$$

where SW_{net} is net shortwave radiation, LW_{down} is incoming longwave, LW_{out} is outgoing longwave, H is sensible heat flux (positive away from surface), LE is latent heat flux, and G is ground heat flux (positive into soil).

The equation is solved by Newton iteration for T_s, using the analytical derivative of the residual. Convergence is typically within 5–8 iterations to 0.01 degC. The resulting T_s is then passed as a **Dirichlet boundary condition** to the subsurface heat solver (Section 2.4). The ground heat flux G in the equation above is not applied as a Neumann BC – it enters implicitly through the Dirichlet temperature. After the heat solver runs, G is recomputed diagnostically from the solved temperature profile (Section 2.5).

4.2 Shortwave radiation

$$SW_{net} = SW_{down} \times (1-\alpha)$$

where α is the surface albedo. Typical values:

Surface type	Albedo	Notes
Bare sand/mudflat (dry)	0.25–0.35	Default: 0.25
Wet bare mudflat	0.10–0.15	Water film darkens surface
Cyanobacterial mat	0.06–0.12	Common in tropical tidal flats
Dark algal biofilm	0.08–0.15	On intertidal mud surfaces

4.3 Longwave radiation

Incoming longwave uses the Brutsaert (1975) clear-sky emissivity with a cloud correction:

$$\varepsilon_{clear} = 1.24\left(\frac{e_a}{T_{air,K}}\right)^{1/7}$$

$$\varepsilon_{sky} = \varepsilon_{clear}\,(1-c)\,+c$$

$$LW_{down} = \varepsilon_{sky}\,\sigma\,T_{air,K}^4$$

where e_a is the atmospheric vapour pressure (kPa), c is cloud cover fraction, and $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4}.

Outgoing longwave:

$$LW_{out} = \varepsilon_s\,\sigma\,T_{s,K}^4$$

where $\varepsilon_s = 0.95$ is the surface emissivity for bare soil.

4.4 Sensible heat flux

Bulk aerodynamic formulation:

$$H = \rho_{air}\,c_p\,C_H\,u\,\left(T_s - T_{air}\right)$$

where C_H is the neutral bulk transfer coefficient:

$$C_H = \frac{\kappa^2}{\left[\ln\left(z_{ref}/z_0\right)\right]^2}$$

Parameter	Symbol	Value	Units
Air density	ρ_{air}	1.2	kg m^{-3}
Specific heat of air	c_p	1004	J kg^{-1} K^{-1}
von Karman constant	κ	0.4	–
Reference height	z_{ref}	10	m
Roughness length	z_0	0.001	m

The reference height is 10 m to match the ERA5 10 m wind product. The roughness length (1 mm) is appropriate for a smooth, bare tidal flat. A minimum effective wind speed of 0.5 m s^{-1} is imposed to prevent singularity in calm conditions.

4.5 Latent heat flux

The latent heat flux is computed from the bulk aerodynamic humidity gradient, scaled by a soil moisture availability factor β:

$$LE = \beta\, \rho_{air}\, L_v\, C_H\, u \frac{0.622}{P}\ \max\left(e_{sat}\left(T_s\right) - e_{air},\, 0\right)$$

where $L_v = 2.45 \times 10^6$ J kg^{-1} is the latent heat of vaporisation, $P = 101325$ Pa is atmospheric pressure, and e_{sat} is the saturation vapour pressure from the Tetens formula:

$$e_{sat}\left(T\right) = 610.78\ \exp\left(\frac{17.27\,T}{T + 237.3}\right) \quad \text{[Pa, T in degC]}$$

The moisture limitation factor is:

$$\beta = \min\left(\frac{\theta_{top}}{\theta_{fc}},\, 1\right)$$

where θ_{top} is the volumetric water content of the surface soil layer and θ_{fc} is the field capacity, defined as $\phi - S_y$ (porosity minus specific yield). When the surface is wet ($\theta \geq \theta_{fc}$), $\beta = 1$ and the surface evaporates at the potential rate. As the surface dries, $\beta \to 0$ and latent heat is suppressed, allowing the surface to heat up under insolation.

The β factor is the critical control on surface temperature in arid-zone intertidal systems. Without it, LE removes ~230 W m^{-2} constantly regardless of soil state, suppressing surface temperatures below 35 degC even under peak summer insolation. With the moisture limitation, exposed dry soil reaches 50–60 degC, consistent with field observations on tropical tidal flats and salt pans.

4.6 Ground heat flux

The ground heat flux couples the SEB to the soil heat equation:

$$G = \frac{\lambda_{top}}{\Delta z_{top}}\left(T_s - T_{soil,0}\right)$$

where λ_{top} is the thermal conductivity of the surface layer (moisture-dependent via the Johansen model) and Δz_{top} is the distance from the surface to the top layer centre (~3 mm for the default 20-layer, 1 m column). A minimum of 3 mm is enforced for numerical stability.

4.7 Newton iteration

The SEB residual is:

$$R\left(T_s\right) = SW_{net} + LW_{down} - LW_{out}\left(T_s\right) - H\left(T_s\right) - LE\left(T_s\right) - G\left(T_s\right) = 0$$

The derivative of the residual with respect to T_s includes contributions from all temperature-dependent terms:

$$\frac{dR}{dT_s} = -4\varepsilon_s \sigma T_{s,K}^3 - \rho c_p C_H u - \beta \rho L_v C_H u \frac{0.622}{P}\frac{de_{sat}}{dT} - \frac{\lambda_{top}}{\Delta z_{top}}$$

where the derivative of e_{sat} is:

$$\frac{de_{sat}}{dT} = e_{sat}(T)\frac{17.27 \times 237.3}{(T + 237.3)^2}$$

The Newton update $T_s^{(k+1)} = T_s^{(k)} - R/\left(dR/dT_s\right)$ is applied iteratively (up to 20 iterations) with convergence criterion $\left|T_s^{(k+1)} - T_s^{(k)}\right| < 0.01$ degC. The result is clamped to $\left[T_{air} - 20,\ T_{max}\right]$ where $T_{max} = 75$ degC for stability.

5. Coupling Between Temperature and Moisture

5.1 Moisture to temperature

Soil moisture affects the thermal solution through three pathways:

1. **Thermal conductivity** (λ): Via the Johansen model. Wet soil conducts heat ~5x faster than dry soil. This determines how rapidly surface forcing penetrates to depth.
2. **Heat capacity** (C): Via the volume-weighted mixture. Wet soil has ~2.5x the heat capacity of dry soil, moderating temperature swings.
3. **Latent heat** (LE): Via the β factor in the SEB. Evaporative cooling is the dominant surface energy sink when the soil is wet.

5.2 Temperature to moisture

The SEB computes the latent heat flux LE, which determines the evaporation rate:

$$E_{actual} = \max\left(\frac{LE}{1000\,L_v},\ 0\right) \times 86400 \quad [\mathrm{m/day}]$$

This evaporation demand is then applied to the unsaturated moisture profile, partitioned between direct water table evaporation and unsaturated-zone extraction. The soil temperature profile also enters the moisture solver's vapour transport term (when enabled).

6. Submerged Boundary Condition

When the water depth exceeds 1 cm, the soil surface is treated as submerged. The surface energy balance is bypassed entirely, and the water temperature from the hydrodynamic model (T_{sea}) is applied directly as a Dirichlet BC to the heat solver.

This is physically appropriate because the hydrodynamic model has already solved its own energy balance on the overlying water column. The soil model receives the result of that calculation and conducts heat into or out of the saturated soil beneath.

At Giralia, sea temperatures of 28–32 degC in summer combined with high thermal conductivity in saturated soil produce efficient deep heating during inundation events. This is the primary mechanism by which heat reaches depth at frequently inundated sites – not solar heating through the surface energy balance.

7. Sensitivity to Surface Properties

7.1 Albedo: bare mudflat vs dark algal mat

Tropical intertidal flats commonly develop surface microbial or algal mats (cyanobacterial crusts). These dark surfaces absorb substantially more shortwave radiation than bare sediment.

Testing at Giralia site 6 (high intertidal, 27% wet fraction, 3-month summer period) showed:

Metric	Bare mudflat ($\alpha = 0.25$)	Algal mat ($\alpha = 0.10$)	Difference
Peak surface temperature	54.6 degC	58.9 degC	+4.3 degC
Mean surface temperature (dry)	28.0 degC	29.1 degC	+1.2 degC
Mean SW absorbed (dry)	222 W m^{-2}	266 W m^{-2}	+44 W m^{-2}
Cumulative ET (3 months)	306 mm	348 mm	+42 mm

The extra 44 W m^{-2} absorbed by the dark surface is partitioned into higher surface temperatures (+4 degC peak) and increased evaporation (+14%). The higher surface temperature drives a stronger vapour pressure gradient, so even though the soil moisture dynamics are similar, total water loss increases. The moisture and water table responses are qualitatively unchanged; the albedo effect propagates primarily through the thermal channel.

Reference albedo values for microbial mats in comparable settings: Shark Bay, Western Australia (0.08–0.10); Arabian Gulf sabkha (0.10–0.12); Baja California tidal flats (0.08–0.15).

7.2 Roughness length

The aerodynamic roughness length $z_0 = 0.001$ m is appropriate for smooth bare tidal flats. Rough surfaces (e.g., mangrove pneumatophore fields, dense crab-mound topography) would have larger z_0 (0.01–0.1 m), which increases the bulk transfer coefficient C_H and strengthens turbulent exchange. This would:

- Increase H (sensible heat loss), reducing surface temperature
- Increase LE (latent heat loss, when $\beta > 0$), further cooling the surface
- Reduce peak surface temperatures by several degrees

This parameter can be adjusted if the model is applied to vegetated or bioturbated zones.

8. Numerical Stability

8.1 Dirichlet vs Neumann surface BC

The surface energy balance (Section 4) computes a surface temperature T_s, which could in principle be applied to the heat solver (Section 2) either as:

- **Dirichlet BC** (used): T_s from the SEB is prescribed at the surface (Section 2.4). The solver computes the subsurface profile consistent with this boundary. G is then computed diagnostically (Section 2.5).
- **Neumann BC** (not used): The ground heat flux G from the SEB would be prescribed as a source term. The solver would compute both the subsurface profile and the surface temperature.

The Neumann approach is more physically direct (it avoids forcing a potentially inconsistent surface temperature) but is numerically unstable with thin surface layers. With the default 20-layer geometric grid, the top layer has $\Delta z \approx 3$ mm, and a Neumann BC produces violent oscillations (temperature swings of hundreds of degrees per timestep).

The Dirichlet approach is unconditionally stable because the implicit solver simply relaxes the profile toward the prescribed surface temperature, regardless of grid spacing. It may introduce a small inconsistency between the SEB's G and the solver's actual surface flux, but in practice this is negligible (< 1 W m^{-2} at convergence).

8.2 Timestep considerations

The implicit backward-Euler scheme is unconditionally stable for any timestep. However, accuracy degrades at very large timesteps because the scheme is only first-order in time. With the standard 1-hour timestep (matching the ERA5 forcing interval), the temperature solution is accurate to within ~0.1 degC for diurnal cycling.

9. Implementation Reference

9.1 Module: `intertidal_soil/temperature.py`

Function	Purpose
`soil_temp(params, vwc, surface_temp, temp, dt, surface_flux)`	One-step heat conduction solve
`initial_temp(params, vwc_init, air_temp, deep_temp, spin_up_days, dt)`	Temperature spin-up to equilibrium
`thomas_solve(a, b, c, d)`	Tridiagonal matrix solver

9.2 Module: `intertidal_soil/surface_energy.py`

Function	Purpose
`surface_energy_balance(sw_down, T_air, rh, wind, T_soil_top, k_soil_top, dz_top, vwc_surface, field_capacity, albedo, emissivity, cloud_cover, max_Ts)`	Newton SEB solve for T_s
`saturation_vapour_pressure(T_C)`	Tetens formula

SEB return values: `Ts`, `G`, `sw_net`, `lw_down`, `lw_out`, `H`, `LE`, `beta`.

9.3 Thermal parameters in `SoilParams`

Parameter	Field	Default	Units
Mineral thermal conductivity	`k_mineral`	2.5	W m^{-1} K^{-1}
Water thermal conductivity	`k_water`	0.57	W m^{-1} K^{-1}
Mineral volumetric heat capacity	`c_mineral`	2.0×10^6	J m^{-3} K^{-1}
Water volumetric heat capacity	`c_water`	4.18×10^6	J m^{-3} K^{-1}
Air volumetric heat capacity	`c_air`	1.25×10^3	J m^{-3} K^{-1}
Mineral density	`mineral_density`	2.6	Mg m^{-3}
Deep temperature	`deep_temp`	26.0	degC

Part C

Giralia Case Study

Part C: Giralia Case Study

C.1 Site Description

Giralia is an arid-zone mangrove system on the eastern shore of Exmouth Gulf, Western Australia (22.1°S, 114.6°E). The tidal flat transitions from subtidal channels through low, mid, and high intertidal zones to supratidal salt flats over a distance of several kilometres. Soil is sandy-clay with macropore structure (bioturbation, crab burrows, root channels) that varies along the inundation gradient: abundant near creeks, negligible on the dry supratidal flats.

The model is applied at 12 monitoring points spanning the full inundation gradient, from permanently submerged subtidal channels to supratidal flats that are inundated only during extreme spring tides.

C.2 Hydrodynamic Forcing

Water depths and water temperatures at the 12 monitoring sites were extracted from a locally calibrated Delft-FM hydrodynamic model of Exmouth Gulf, providing hourly output from 2016-01-01 to 2020-12-31. The hydrodynamic model resolves the tidal dynamics of the gulf including the complex wetting and drying of the intertidal flats.

The 12 sites span a wide inundation gradient:

Category	Sites	Wet fraction	f_{macro}
Subtidal	4, 12	79–100%	0.030
Low intertidal	5, 7	49–56%	0.025
Mid intertidal	1, 3	32–33%	0.015
High intertidal	6	20%	0.008
Supratidal	2, 8, 9, 10, 11	5–10%	0.002

When submerged, the Delft-FM water temperature is applied directly as a Dirichlet boundary condition to the soil heat solver. The hydrodynamic model has already solved the water column energy balance; the soil model receives the result.

C.3 Meteorological Forcing

Hourly ERA5 reanalysis data were obtained from the Open-Meteo archive API for the period 2016–2020 at coordinates (-22.1, 114.6). Variables used:

- `temperature_2m` (°C)
- `relative_humidity_2m` (%)
- `wind_speed_10m` (km/h – converted to m/s by dividing by 3.6)
- `shortwave_radiation` (W m^{-2})
- `precipitation` (mm – converted to m/day)
- `cloud_cover` (% – converted to fraction 0–1)

The reference height of 10 m matches the ERA5 10 m wind product.

C.4 Soil Parameters

Base parameters for the sandy-clay tidal flat:

Parameter	Value	Units	Notes
`n_layers`	20	–	Geometric spacing, 1 m column
ψ_e	-1.5	J kg^{-1}	Sandy-clay air entry
K_s	2×10^{-4}	kg s m^{-3}	Moderate-low matrix conductivity
b	4.0	–	Sandy-clay retention shape
ρ_b	1.55	Mg m^{-3}	Compacted tidal flat
S_y	0.15	–	Specific yield
T_{deep}	26.0	°C	Stable tropical deep temperature
α	0.25	–	Bare mudflat (0.10 for algal mat)

Macropore parameters vary along the inundation gradient, reflecting the observation that bioturbation (crab burrows, root channels) is most intense near tidal creeks and negligible on the dry supratidal flats:

Category	f_{macro}	k_{macro} (m/s)	$\alpha_{exchange}$ (s^{-1})
Subtidal	0.030	0.010	5×10^{-5}
Low intertidal	0.025	0.010	5×10^{-5}
Mid intertidal	0.015	0.008	4×10^{-5}
High intertidal	0.008	0.005	3×10^{-5}
Supratidal	0.002	0.002	1×10^{-5}

C.5 Callable Forward-Step Interface

The model is designed for integration into a larger coupled system. The `GiraliaSoilColumn` class wraps the intertidal soil model with a callable `.step()` method that advances one timestep given external forcing:

```
state, diag = col.step(
    water_depth,      # m (0 = dry)
    water_temp,       # degC (from Delft-FM)
    air_temp,         # degC
    sw_down,          # W/m2
    rh,               # %
    wind,             # m/s
    rain,             # m/day
    cloud_cover,      # fraction 0-1
    albedo,           # surface albedo (can vary per step)
)
```

The step method tests submergence (`water_depth > 0.01` m). If submerged, `water_temp` is applied as a Dirichlet BC and the moisture solver runs with ponding. If exposed, the surface energy balance computes T_s using the supplied `albedo`, which is applied as the Dirichlet BC, and the moisture solver runs with ET derived from the SEB latent heat flux. Albedo is a per-step argument (not a column property) so that the master model can vary it over time as surface conditions change (e.g. algal mat development).

C.6 Surface Energy Balance Configuration

Parameter	Value	Notes
Albedo α	0.25 (bare) / 0.10 (algal mat)	
Emissivity ε_s	0.95	Bare soil
Roughness z_0	0.001 m	Smooth tidal flat
Reference height z_{ref}	10 m	ERA5 wind
Longwave down	Brutsaert (1975) + cloud correction	
Latent heat	Bulk aerodynamic, moisture-limited	
β	$\min\left(\theta_{top}/\theta_{fc}, 1\right)$	
θ_{fc}	$\phi - S_y = 0.254$	

The β factor is essential: without it, LE removes approximately 230 W m^{-2} constantly regardless of soil moisture, suppressing surface temperatures below 35 °C. With the moisture limitation, dry exposed soil reaches 50–60 °C under peak summer insolation, consistent with field observations on tropical salt flats.

C.7 Results: 12-Month Multi-Site Simulation

All 12 sites were run for 12 months from 2017-01-01 with hourly timesteps.

Site	Category	Wet %	ET (mm)	WT end (cm)	Mean T_s
1	Mid intertidal	38	1346	0	22.2
2	Supratidal	6	1557	15	22.4
3	Mid intertidal	36	1464	0	22.4
4	Subtidal	100	0	0	23.0
5	Low intertidal	52	1066	0	22.0
6	High intertidal	27	1142	50	22.5
7	Low intertidal	62	828	0	22.2
8	Supratidal	8	1843	0	21.9
9	Supratidal	11	1839	0	21.9
10	Supratidal	10	1641	14	21.8
11	Supratidal	8	1567	10	22.1
12	Subtidal	80	451	0	23.0

Key patterns:

- **Subtidal** (4, 12): Zero ET (always submerged), surface temperature tracks sea temperature.
- **Low intertidal** (5, 7): Frequent tidal recharge keeps the water table shallow, moderate ET (830–1070 mm/yr).
- **Mid intertidal** (1, 3): Water table oscillates with spring/neap tidal cycles, 1350–1460 mm/yr ET.
- **High intertidal** (6): Deepest sustained water table drawdown (50 cm by year-end), 1140 mm/yr.
- **Supratidal** (2, 8–11): Sharp water table drawdown between infrequent tidal events, 1560–1840 mm/yr ET, surface reaches 50+ degrees during dry spells.

Annual ET of 1500–1800 mm for supratidal sites is approximately 50–60% of pan evaporation at Exmouth (approx. 3000 mm/yr), consistent with bare soil over a shallow water table.

C.8 Albedo Sensitivity: Bare Mudflat vs Dark Algal Mat

Tropical intertidal flats commonly develop surface microbial or algal mats (cyanobacterial crusts). These dark surfaces absorb substantially more shortwave radiation than bare sediment. Reference albedo values for microbial mats in comparable settings: Shark Bay, WA (0.08–0.10); Arabian Gulf sabkha (0.10–0.12); Baja California tidal flats (0.08–0.15).

A comparison at site 6 (high intertidal, 3-month summer period):

Metric	Bare (a=0.25)	Algal mat (a=0.10)	Difference
Peak surface temp	54.6 degC	58.9 degC	+4.3 degC
Mean surface temp (dry)	28.0 degC	29.1 degC	+1.2 degC

Metric	Bare (a=0.25)	Algal mat (a=0.10)	Difference
Mean SW absorbed (dry)	222 W/m2	266 W/m2	+44 W/m2
Cumulative ET (3 months)	306 mm	348 mm	+42 mm

The extra 44 W/m2 absorbed by the dark surface partitions into higher peak surface temperatures (+4.3 degC) and increased evaporation (+14%). The warmer surface drives a stronger vapour pressure gradient. Moisture and water table dynamics are qualitatively unchanged; the albedo effect propagates primarily through the thermal channel.

Giralia Case Study Figures

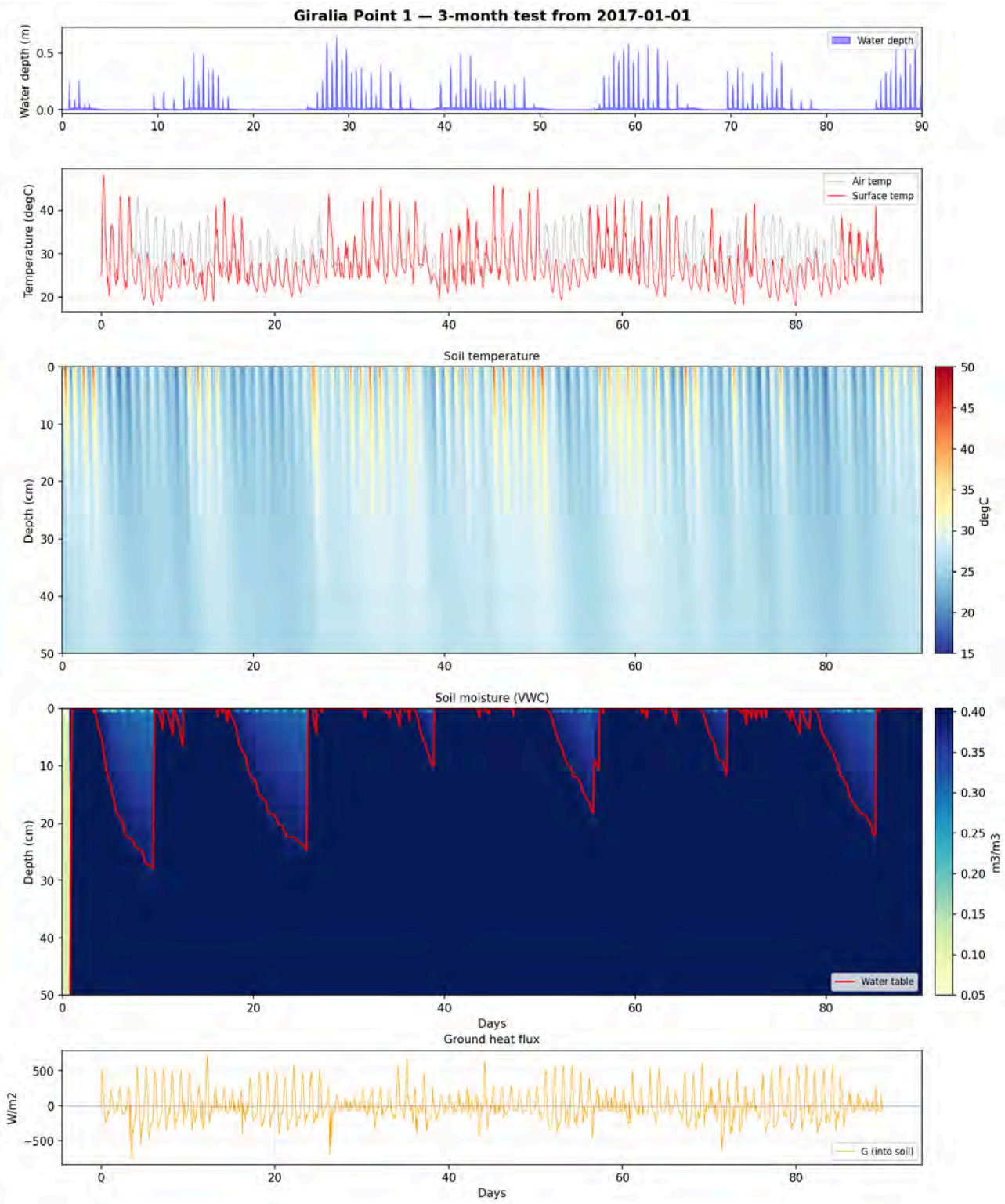

Figure C.1: Site 1 (mid-intertidal, 48% wet), 3-month summer simulation from 2017-01-01.

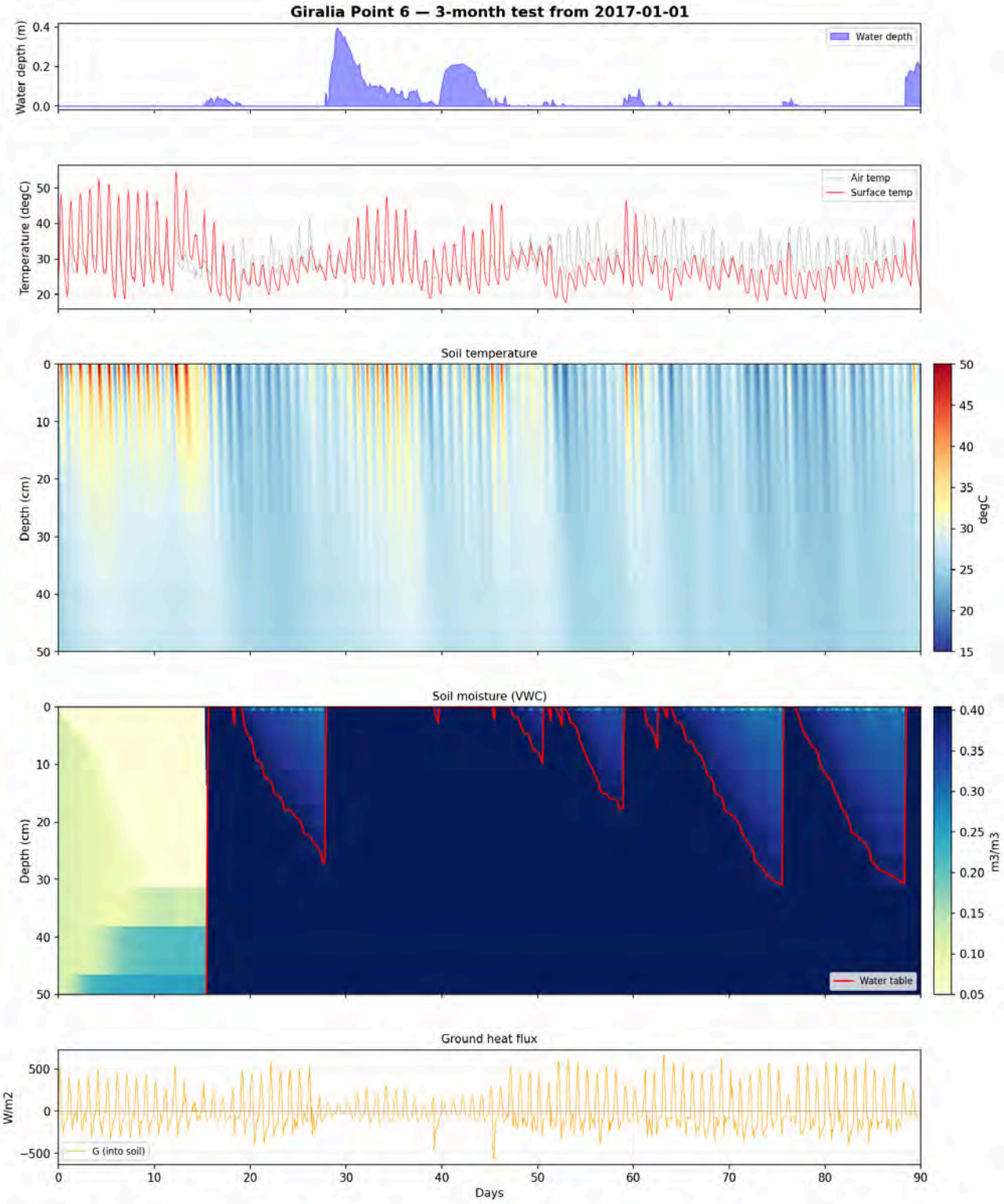

Figure C.2: Site 6 (high-intertidal, 32% wet), 3-month summer simulation from 2017-01-01.

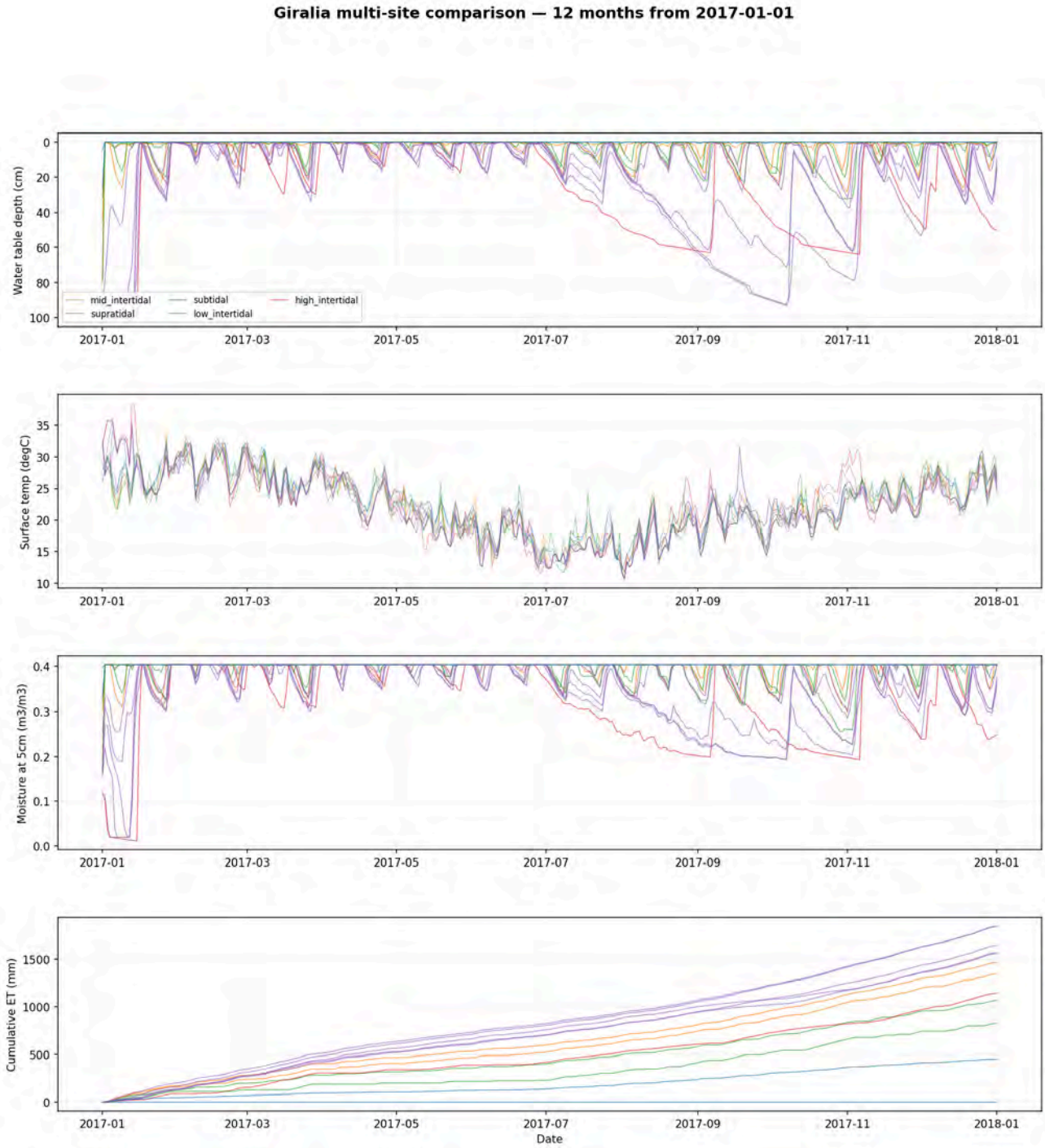

Figure C.3: Multi-site comparison across 12 sites, 12 months from 2017-01-01.

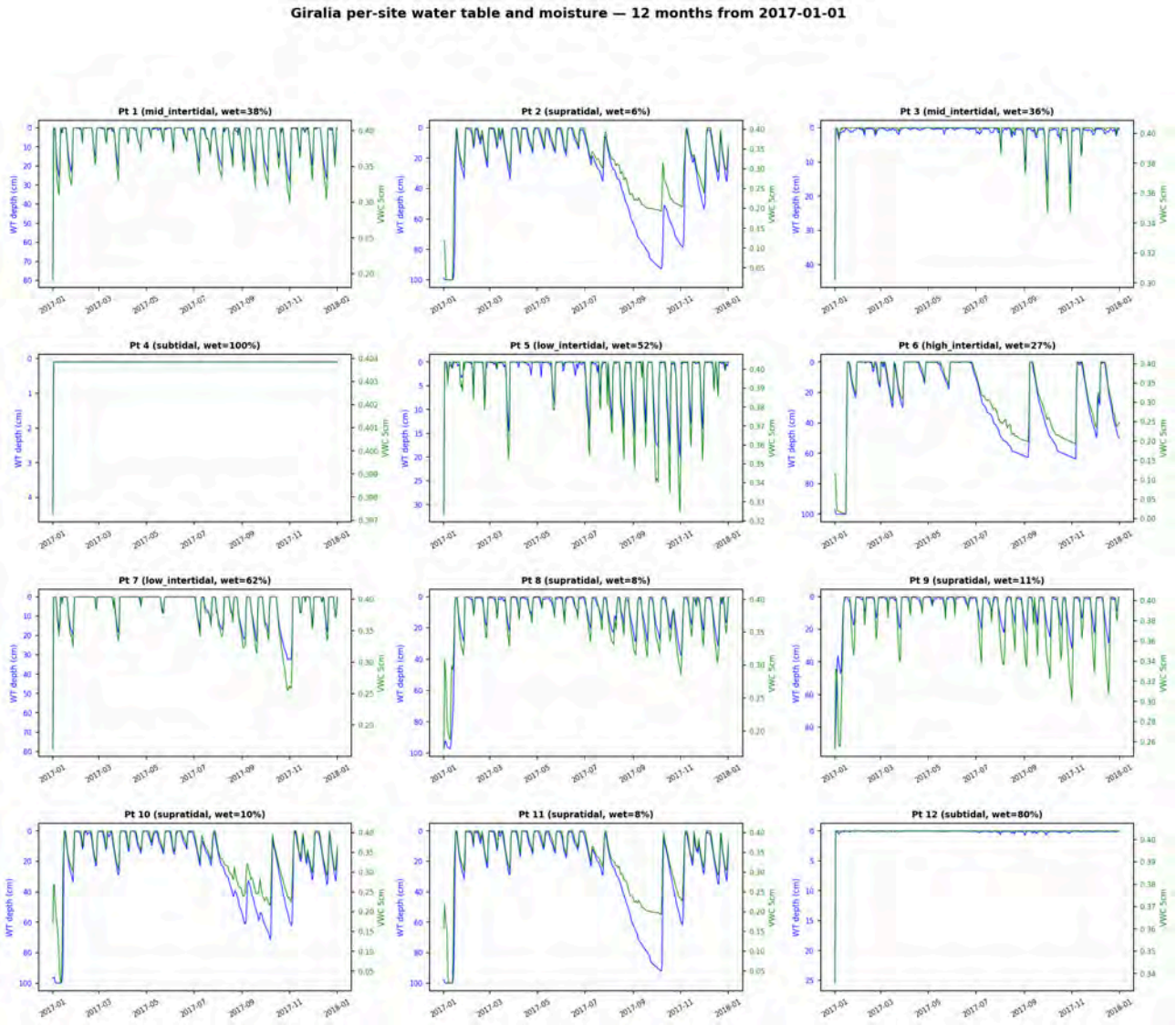

Figure C.4: Per-site water table and moisture, 12 months from 2017-01-01.

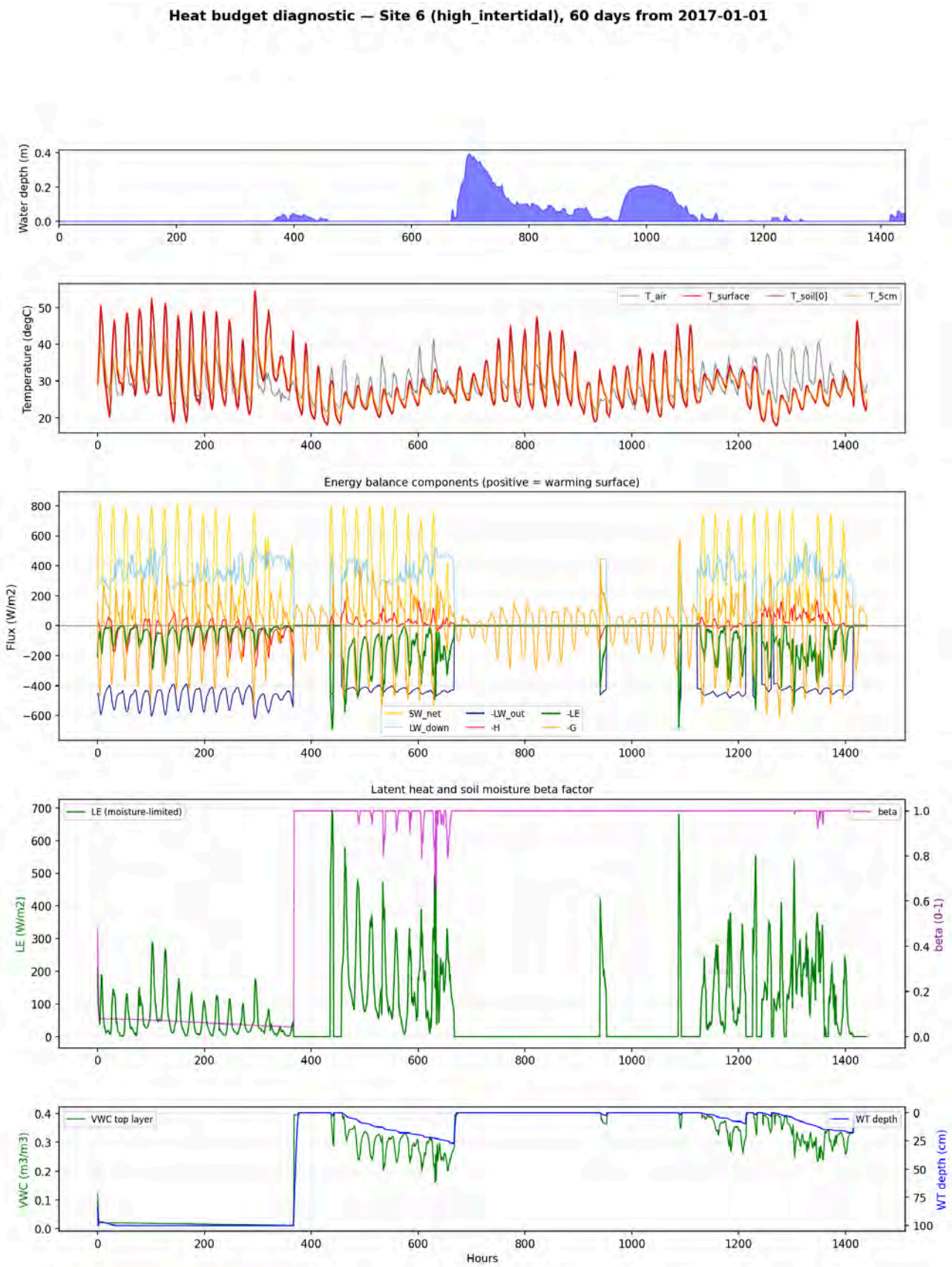

Figure C.5: Surface energy balance diagnostic, Site 6 (high-intertidal), 60 days summer.

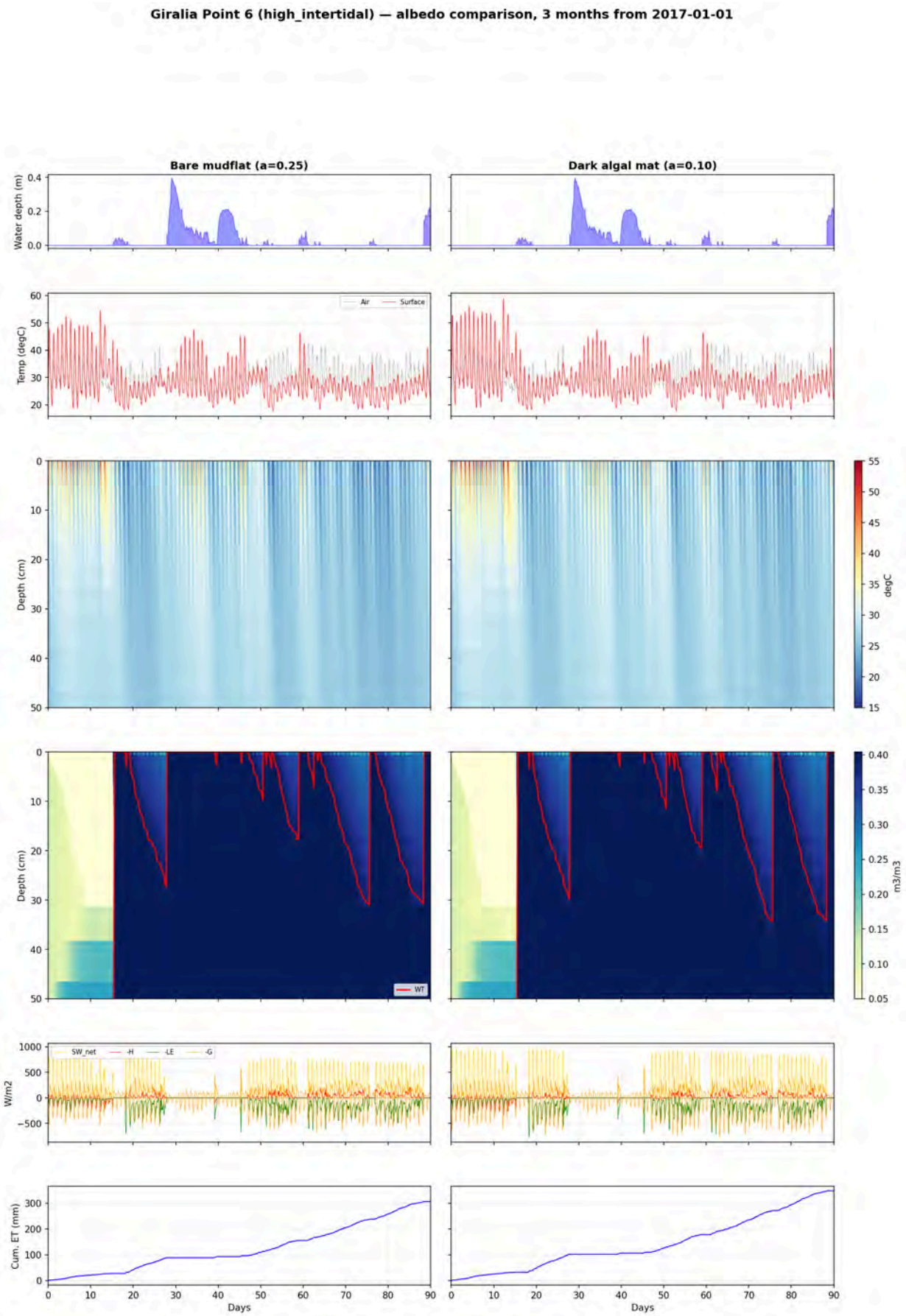

Figure C.6: Albedo comparison at Site 6 -- bare mudflat (0.25) vs dark algal mat (0.10).